A SMALLER LIFT BOX IN THE DEGREE-FIVE HEILBRONN CONSTRUCTION

Abstract. We prove a lower bound $cn^{-2+\eta}$ for the least ordinary triangle area of an n-point set in the unit square, for every integer $n \geq N_0$, with $\eta = 2/100866719$, $c = 2^{-10086671700}$, and $N_0 = 2^{100866718 \cdot 2^{50}}$. The proof retains OpenAI's projective lattice construction and Blinov's degree-five norm decomposition, reserved label tags, and slab cap. Its changed scale is $N = (hq)^4/B^{102195}$. We include the digit moment estimate, singular determinant count, deletion, and effective transfer to every sufficiently large cardinality. The exponent satisfies $\eta > 1/50893237$.

1. Statement, scope, and attribution

For a finite set $P \subset [0,1]^2$ with at least three points, let $\Delta(P)$ be the least ordinary Euclidean area of a triangle formed by three distinct points of P. Thus area includes the factor $1/2$. The main result is the following.

Theorem 1. *Put*

$$\eta = \frac{2}{100866719}, \qquad c = 2^{-10086671700}, \qquad N_0 = 2^{100866718 \cdot 2^{50}}.$$

For every integer $n \geq N_0$ there is a set of n distinct points $P \subset [0,1]^2$ such that $\Delta(P) \geq cn^{-2+\eta}$. All choices in the construction can be made by finite exhaustive search. Moreover, $\eta > 1/50893237$.

The projective sampling, lattice and orbit estimates, treatment of zero determinants, and deletion framework originate in OpenAI's September 25, 2026 manuscript [4]. Blinov's pinned checkpoint [2] supplies the degree-five decomposition, reserved tag words, fresh-digit collision argument, and slab refinement. The paired eighteen-form formula comes from the determinant decomposition of Conner, Gesmundo, Landsberg, and Ventura [3, Theorem 4.1]. Their theorem states a complex Waring decomposition; the precise finite-field identity needed here is verified coefficientwise below. The sole external analytic theorem used is the explicit prime-counting estimate of Bennett, Martin, O'Bryant, and Rechnitzer [1, Theorem 1.2 and (1.10)–(1.11)]. The remaining estimates are proved here with the constants of the recorded proof.

The refined lift scale is $N = (hq)^4/B^{102195}$. The following sections prove the digit moments, singular determinant estimates, deletion and transfer to every sufficiently large cardinality, with all parameters specified. The exact exponent comparison is

$$100866719 < 2 \cdot 50893237 = 101786474.$$

2. Degree-five norm digits and the collision moment

Fix $k = 408781$, a prime $r \geq 37$ with $r \equiv 1 \pmod 3$, and set

$$\begin{aligned} M = 400k^2r^3, \qquad L_1 = L_2 = r, \qquad L_3 = rM, \\ 40000k^4r^6 < B \leq 80000k^4r^6, \qquad h = B^k, \qquad \tau = \left\lfloor B^{k-1}/2 \right\rfloor, \end{aligned} \tag{1}$$

where B is prime. This section proves, for the digit matrix C defined below and its smaller Smith divisor D, that

$$\mathbb{E}[D\mathbf{1}_{\mathcal{J}}] \leq \frac{6}{Mr^5} + \frac{2}{r^{10}}, \tag{2}$$

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where $\mathcal{J}$ is the event that $\det C \pmod h$ has an integer representative of absolute value at most τ.

2.1. **The norm decomposition.** Let $F = \mathbb{F}_r$, $K = \mathbb{F}_{r^5}$, and choose an F-basis $\beta_0, \dots, \beta_4$ of K. For a group X of fifteen variables over F, put $Z_i(X) = \sum_{s=0}^{4} \beta_s X_{i,s}$, $i = 1, 2, 3$. Let σ raise coefficients to their rth powers while fixing variables, and define

$$\mathcal{D} = \det\bigl(Z_i(X^{(j)})\bigr)_{i,j=1}^{3}, \qquad \mathcal{N} = \prod_{a=0}^{4} \sigma^a(\mathcal{D}).$$

Frobenius permutes the five factors, so $\mathcal{N}$ has coefficients in F; it is homogeneous of degree five in each variable group. At F-valued variables it equals the field norm of $\mathcal{D}$, and is nonzero when $\mathcal{D} \neq 0$.

An element $z \in F^*$ of order three exists by Cauchy's theorem. Set $u = -z$ and $v = 1 - u$, so $u^2 - u + 1 = 0$. For each permutation p and coefficient triple c in the following table, put $h_{p,c}(x, y) = \sum_{i=1}^{3} c_i x_{p(i)} y_i$.

p	First triple	Second triple	Third triple
$(1,2,3)$	$(-u,-1,v)$	$(-v,-1,u)$	$(-v,v,v)$
$(1,3,2)$	$(-1,-u,-v)$	$(v,-u,1)$	$(u,-u,-u)$
$(2,1,3)$	$(-u,v,1)$	$(-v,v,-v)$	$(-v,u,1)$
$(3,1,2)$	$(-1,-u,v)$	$(-1,-v,u)$	$(-1,1,-1)$
$(2,3,1)$	$(-1,-1,1)$	$(v,u,1)$	$(u,v,1)$
$(3,2,1)$	$(1,-u,v)$	$(-v,-v,v)$	$(1,-v,u)$

Writing the eighteen forms as h_t, their identity is

$$\sum_{t=1}^{18} \prod_{j=1}^{3} h_t(A_{\cdot j}, C_{\cdot j}) = 3 \det A \det C. \tag{3}$$

Here is a finite coefficient verification in $\mathbb{Z}[u]/(u^2 - u + 1)$. For $a, b \in \{1,2,3\}^3$, the coefficient of $\prod_j A_{a_j,j} C_{b_j,j}$ on the left is the sum of $\prod_j c_{b_j}$ over the listed forms satisfying $a_j = p(b_j)$ for all j. Reducing u^2 to $u - 1$ gives zero unless both triples are permutations; otherwise it gives $3\,\mathrm{sign}(a)\,\mathrm{sign}(b)$. These are precisely the coefficients on the right. The packet's checker implements this calculation by

$$(a, b)(c, d) = (ac - bd, ad + bc + bd),$$

representing $a + bu$. It expands $18 \cdot 27 = 486$ terms and verifies the thirty-six nonzero coefficients.

Apply (3) to the pairs $(\sigma Z, \sigma^2 Z)$ and $(\sigma^3 Z, \sigma^4 Z)$. For $w = (t_1, t_2) \in \{1, \dots, 18\}^2$ set

$$H_w(X) = h_{t_1}(\sigma Z(X), \sigma^2 Z(X)) h_{t_2}(\sigma^3 Z(X), \sigma^4 Z(X)), \qquad F_{i,w}(X) = Z_i(X) H_w(X).$$

Multiplying the two identities and $\det Z$ yields

$$9\mathcal{N} = \sum_{w} \det\bigl(F_{i,w}(X^{(j)})\bigr). \tag{4}$$

Each entry is homogeneous of degree five and has coefficients in K.

Choose $\theta \in K \setminus F$. Its degree over F divides five and is not one, so $1, \theta, \dots, \theta^4$ is a basis. Let $\ell : K \to F$ take the constant coordinate in this basis. Write the coefficients of $F_{i,w}$ in this basis to obtain $P_{i,w}(Y, X) \in F[Y, X]$ of Y-degree at most four and with $P_{i,w}(\theta, X) = F_{i,w}(X)$. For $s = 0, \dots, 8$ let

$$L_s(Y) = \prod_{\substack{0 \le t \le 8 \\ t \neq s}} \frac{Y - t}{s - t}.$$

The nine nodes are distinct. If $\widehat{x}, \widehat{y}$ are the degree-at-most-four representatives of $x, y \in K$, interpolation gives

$$\ell(xyz) = \sum_{s=0}^{8} \widehat{x}(s)\widehat{y}(s)\ell(L_s(\theta)z). \tag{5}$$

Indeed, interpolate $\widehat{x}\widehat{y}$ in degree at most eight, evaluate at θ, multiply by z, and apply ℓ. The identity also holds coefficientwise for polynomials.

Expand each determinant in (4) and apply (5). Since $\ell(\mathcal{N}) = \mathcal{N}$, the polynomials

$$\begin{aligned} f_{1,w,s}(X) &= 9^{-1}P_{1,w}(s, X), \\ f_{2,w,s}(X) &= P_{2,w}(s, X), \\ f_{3,w,s}(X) &= \ell(L_s(\theta)F_{3,w}(X)) \end{aligned}$$

are F-valued, homogeneous of degree five, and satisfy

$$\mathcal{N} = \sum_{w,s} \det(f_{i,w,s}(X^{(j)})). \tag{6}$$

There are $T = 9 \cdot 18^2 = 2916$ terms. All inverses used exist since $r \geq 37$. For $\xi \in K$ let $X(\xi)$ be the coordinates of $(1, \xi, \xi^2)$. At three distinct labels the determinant before taking the norm is the nonzero Vandermonde product $\prod_{i<j}(\xi_j - \xi_i)$; hence (6) is nonzero.

2.2. **Positions, reserved tags, and digits.** Let $\mathcal{C}$ consist of the integers with twelve ternary digits, each zero or one. If $x + y = 2z$ in $\mathcal{C}$, neither side has ternary carries, so $x_i + y_i = 2z_i$ in each digit and $x = y = z$. The order-preserving conversion of a binary integer to ternary digits zero or one takes 2920, whose binary expansion is 101101101000, to 204390. Thus exactly 2921 words are at most 204390. The words below ten are $0, 1, 3, 4, 9$. Consequently

$$\mathcal{A} = \mathcal{C} \cap [10, 204391)$$

has 2916 elements. Enumerate them as u_a, $0 \leq a < 2916$, and match them bijectively with the terms of (6). Use row positions

$$i_a = j_a = u_a, \qquad l_a = k - 1 - 2u_a.$$

They are injective within each row and lie in $[0, k)$, including $l_a = 0$ at the largest word. Further, $i_a + j_b + l_c = k - 1$ is equivalent to $u_a + u_b = 2u_c$, so holds exactly when $a = b = c$.

Write $\xi = \sum_s \lambda_s \beta_s$ and let $(t_0, \ldots, t_4) = (0, 1, 3, 4, 9)$. Prescribe residue one in row one, position zero; prescribe λ_s in row two, position t_s. At each norm position in row i prescribe $f_{i,a}(X(\xi))$, and prescribe zero modulo r elsewhere. There is no conflict within a row. A combination contributing at the target coefficient and involving a tag would give $x + y = 2z$ with $x, y \in \mathcal{A} \cup \{0, 1, 3, 4, 9\}$ and $z \in \mathcal{A}$. Midpoint-freeness forces equality of all three words and excludes tags.

Conditional on the label, choose independently at every position in row i a uniform positive digit in $[1, L_i]$ with its prescribed residue. The top two rows have one choice per position, and every bottom digit has M choices. Each column uses a fresh independent uniform label and fresh digit draws. Put $c_i = \sum_{v=0}^{k-1} d_{i,v}B^v \pmod h$. Every digit is positive and less than B, so every entry is a unit modulo h.

For three columns write

$$F(Y) = \det\left(\sum_{v=0}^{k-1} d_{i,j,v}Y^v\right) = \sum_e F_e Y^e, \qquad P = L_1L_2L_3 = 400k^2r^6.$$

Each coefficient has at most k^2 position choices per determinant permutation. Hence $|F_e| \leq A_0 = 6k^2P$, and matching gives

$$F_{k-1} \pmod r = \mathcal{N}(X(\xi_1), X(\xi_2), X(\xi_3)). \tag{7}$$

Since $B-1 \geq 100k^2P$, $A_0/(B-1) \leq 3/50$. The integer $S = \sum_{e=0}^{k-1} F_e B^e$ represents $\det C \pmod h$ and satisfies $|S| < h/2$. Its terms below degree $k-1$ have total absolute value at most $\frac{3}{50}(B^{k-1}-1)$. If $F_{k-1} \neq 0$ as an integer, then

$$|S| \geq (47B^{k-1}+3)/50 > \tau.$$

All other representatives $S + mh$ have absolute value greater than $h/2$. Thus $\mathcal{J}$ implies $F_{k-1} = 0$ over the integers, and (7) excludes $\mathcal{J}$ for distinct labels.

2.3. **Smith divisor tails.** Let b be the largest $j \leq k$ for which every two-by-two minor of C is divisible by B^j, and set $D = B^b$. A unit pivot clears a row and column. Pivoting again on an entry of least valuation in the remaining block gives $\operatorname{diag}(1, B^b, B^e)$ with $b \leq e \leq k$; valuation k denotes zero. Invertible operations preserve the ideal of two-by-two minors, proving this characterization even for $j = k$.

Fix any labels and condition on the first column. The top two rows are deterministic. Since C_{11} is a unit, $b \geq j$ implies

$$C_{3v} = C_{31}C_{11}^{-1}C_{1v} \pmod{B^j}, \qquad v = 2, 3.$$

Each congruence determines the first j base-B digits of a bottom entry. Independence and M choices per digit give $\mathbb{P}(b \geq j \mid \text{labels}) \leq M^{-2j}$. As $\theta_0 = B/M^2 \leq 1/2$, the tail identity

$$B^b = 1 + \sum_{j=1}^{k} (B^j - B^{j-1})\mathbf{1}_{\{b \geq j\}}$$

implies $\mathbb{E}[D \mid \text{labels}] \leq 1 + \sum_{j \geq 1} \theta_0^j \leq 2$.

2.4. **Exactly two equal labels and a fresh high digit.** Permute columns to labels (λ, λ, μ) with $\lambda \neq \mu$. This preserves minor divisibilities and $\mathcal{J}$. Write the deterministic top polynomials as $A_\lambda, Q_\lambda, A_\mu, Q_\mu$ and the bottom polynomials as Z_1, Z_2, Z_3. Subtracting the first column from the second gives

$$F(Y) = U(Y)(Z_1(Y) - Z_2(Y)), \qquad U = A_\lambda Q_\mu - A_\mu Q_\lambda.$$

Let s be the first differing coordinate of the labels and put $a = t_s$. Below degree ten the A polynomials have digits $1, r, r, \ldots$; the Q polynomials agree below a and differ at a by a nonzero integer of absolute value less than r. Thus U vanishes below a and $0 < |U_a| < r < B$. The top minor $U(B)$ has exact B-adic valuation a, so $b \leq a < 10$.

Put $v = k - 1 - a > a$. Holding all other digits fixed, F_{k-1} is linear in the bottom digit of column two at v, with nonzero coefficient $-U_a$. At most one of its M values gives zero, whence $\mathbb{P}(\mathcal{J} \mid \text{labels}) \leq 1/M$. For $1 \leq j \leq a$, condition on the first column and all digits of bottom columns two and three at positions at least j, except that digit at v. The two Schur congruences determine the low prefixes at cost at most M^{-2j}. The unused digit remains independent and uniform, and the coefficient equation costs at most $1/M$. Since $v > a \geq j$,

$$\mathbb{P}(b \geq j, \mathcal{J} \mid \text{labels}) \leq M^{-2j-1}.$$

The truncated tail identity gives

$$\mathbb{E}[D\mathbf{1}_{\mathcal{J}} \mid \text{labels}] \leq M^{-1}\left(1 + \sum_{j=1}^{a} (B/M^2)^j\right) \leq 2/M. \tag{8}$$

Independent uniform labels range over r^5 possibilities. Exactly two equal labels have probability at most $3/r^5$, and three equal labels have probability $1/r^{10}$. Use (8) in the first case and the uniform bound two in the second. Three distinct labels cannot give $\mathcal{J}$. Averaging proves (2), with all label patterns included.

3. Parameters and geometric counting

3.1. **Explicit parameters and primes.** Use the digit distribution of Section 2, and put

$$k = 408781, \quad s_0 = 102195, \quad \rho = 249k + 3 - 9s_0 = 100866717, \qquad \gamma = \frac{\rho}{\rho+1}, \tag{9}$$
$$\alpha = \rho + \gamma, \quad \eta = \frac{2}{\rho+2}, \quad R = 2^{2^{50}}, \quad N_0 = 2^{(\rho+1)2^{50}}, \quad c = 2^{-100\rho}.$$

The unconditional theorem of Bennett, Martin, O'Bryant, and Rechnitzer [1, Theorem 1.2 and (1.10)–(1.11), pp. 4–5 of arXiv v3] implies that, for coprime a, m, $3 \le m \le 10^5$, and $x \ge 8 \cdot 10^9$,

$$\left|\vartheta(x;m,a) - \frac{x}{\varphi(m)}\right| < \frac{x}{160\log x}, \qquad \vartheta(x;m,a) = \sum_{\substack{p \le x \text{ prime} \\ p \equiv a \pmod m}} \log p. \tag{B1}$$

Here log is natural; $\log_2$ below is base two. For $m = 3$ the displayed bound is weaker than the source's $1/840$ bound. Apply it at x and $2x$ with $a = 1$ to get

$$\vartheta(2x;3,1) - \vartheta(x;3,1) > \frac{x}{2} - \frac{2x}{160\log(2x)} - \frac{x}{160\log x} \ge x\left(\frac{1}{2} - \frac{3}{160\log x}\right) > 0.$$

Thus $(x, 2x]$ contains a prime congruent to one modulo three whenever $x \ge 8 \cdot 10^9$; it also supplies ordinary primes when required.

For each prime $r \ge R$, $r \equiv 1 \pmod 3$, choose B in the interval (1), then a prime $h^6 < q \le 2h^6$. Both interval endpoints exceed the analytic threshold. The primes B, r, q are distinct and odd. Set

$$H = 2h, \qquad N = \frac{(hq)^4}{B^{s_0}} = B^{4k-s_0}q^4.$$

This is an integer and $N/(hq) = B^{3k-s_0}q^3$ is a positive integer. We have

$$N \ge h, \quad \tau \ge B^{k-1}/3, \quad 2\tau < h, \quad \tau < h < q, \quad h \ge 10, \quad q > h^6 \ge 2000H^2.$$

In particular the subsequent estimates use this smaller N, and never assume $N = (hq)^4$.

3.2. **Elementary lattice bounds.** Lengths and covolumes are Euclidean. In a rank-s lattice L, $s \le 3$, choose $v_1, \dots, v_s$ successively shortest outside the preceding real span and write $\lambda_i = |v_i|$. Then

$$\det L \le \prod_i \lambda_i \le s^{s/2} \det L. \tag{B2}$$

The first inequality follows by taking the sublattice spanned by the v_i. For the second, use orthonormal coordinates aligned with their successive spans. The open box $|x_i| < \lambda_i/\sqrt{s}$ has no nonzero lattice point: its last nonzero coordinate j would produce a vector outside the preceding span with length less than λ_j. The translates by L of half this box have disjoint interiors. Integration over a fundamental domain gives $s^{-s/2}\prod_i \lambda_i \le \det L$. In rank three the successive vectors therefore span a sublattice of index less than six.

In rank two, with covolume J and successive lengths λ_1, λ_2, any translate has at most $32T^2/J$ points in a ball of radius $T \ge \lambda_2$. Indeed, the parallelograms of the span of the two successive vectors have diameter at most $2T$. For each coset, the parallelograms attached to points in the ball are disjoint and lie in a disk of radius $3T$. Their area bounds the count by $9\pi T^2/\det L_0$; summing the cosets gives $9\pi T^2/J < 32T^2/J$. If $T < \lambda_2$, points in the centered ball lie on the shortest-vector line and number at most $1 + 2T/\lambda_1$.

For primitive $y \in \mathbb{Z}^3$, $L_y = \mathbb{Z}^3 \cap y^\perp$ has covolume $|y|$. More generally, if $\Lambda \subset \mathbb{Z}^3$ has index I and $y \cdot \Lambda = g\mathbb{Z}$, then $\Lambda \cap y^\perp$ has covolume $I|y|/g$. Choose $u \in \Lambda$ with $y \cdot u = g$. A basis of the kernel together with u is a basis of Λ, and its height over the kernel plane is $g/|y|$. Primitivity gives $g = 1$ for $\mathbb{Z}^3$.

For $R_1 \ge R_2 \ge R_3 \ge 1$ and a fixed nonzero integer v, the number of integer column triples with $|u_i| \le R_i$ and determinant v is at most

$$2^{33}(R_1R_2R_3)^2\bigl(\log_2(2R_1)\bigr)^2. \tag{B3}$$

To prove this, the independent columns u_2, u_3 have cross product a nonzero integer multiple of a primitive normal y, with $|y| \le R_2R_3$. If μ_1, μ_2 are the successive lengths of L_y, then $\mu_1 \le R_3$ and $\mu_2 \le R_2$. For fixed u_2, u_3, the first column lies in a translate of L_y, giving at most $32R_1^2/|y|$ choices. The second has at most $32R_2^2/|y|$ choices. If $\mu_2 \le R_3$, the third has at most $32R_3^2/|y|$ choices. In $Y \le |y| < 2Y$, $Y \ge 1$, there are at most $125Y^3$ integer vectors. Hence the sum of $|y|^{-3}$ up to R_2R_3 is at most $250\log_2(2R_1)$, and this part contributes at most $2^{23}(R_1R_2R_3)^2\log_2(2R_1)$.

If $\mu_2 > R_3$, group normals by dyadic $U \le \mu_1 < 2U$, $Z \le \mu_2 < 2Z$. By (B2), $UZ/2 \le |y| < 4UZ$. A shortest primitive $z \in L_y$ has $U \le |z| < 2U$, giving at most $125U^3$ choices. Normals for fixed z belong to L_z and have length less than $4UZ$. The second successive length of L_z is at most $2|z| < 4U$, since the first is at least one and (B2) applies. The plane count gives at most $32(4UZ)^2/|z| \le 512UZ^2$ choices for y. Thus there are fewer than $2^{16}U^4Z^2$ normals in the group. For each, the first two column counts are at most $64R_1^2/(UZ)$ and $64R_2^2/(UZ)$, while the third lies on the shortest line and has at most $3R_3/U$ choices. One group contributes at most $2^{30}R_1^2R_2^2R_3U \le 2^{30}(R_1R_2R_3)^2$. There are at most $(\log_2(2R_1))^2$ groups. Combining the two cases proves (B3).

Consequently, if $\Lambda \subset \mathbb{Z}^3$ has index I and $\lambda_3 \le X$, $X \ge 2$, the number of matrices with rows in Λ, row lengths at most X, and a fixed nonzero determinant is at most

$$2^{80}\bigl(\log_2(2X)\bigr)^2X^6/I^2. \tag{B4}$$

Let V have the successive shortest vectors as rows, and let $m = [\Lambda : \mathbb{Z}^3V] < 6$. Then $Q = mAV^{-1}$ is integral, and the map $A \mapsto Q$ is injective. Cramer's rule and (B2) bound a coordinate relative to V by $6X/\lambda_j$ in absolute value. Multiplying by m and taking column length bounds column j of Q by $64X/\lambda_j$. These radii lie between one and $64X$ and have product at most $64^3X^3/I$. The determinant of Q is fixed and nonzero; if it is not integral the count is zero. Apply (B3). Since $\log_2(128X) \le 7\log_2(2X)$, the resulting constant is at most $2^{33+36+6} = 2^{75} < 2^{80}$.

3.3. **Row lattices and special-linear orbits.** A matrix C over $\mathbb{Z}/h\mathbb{Z}$ with a unit entry has Smith form $\operatorname{diag}(1, D, E)$, where $D = B^b$, $E = B^e$, and $0 \le b \le e \le k$, including zeros represented by B^k. The pivot and minor-ideal argument was given in Section 2. Let Λ be the inverse image of the row span of C modulo h. The diagonal module has size $h^3/(DE)$, and column operations preserve module sizes, so

$$I = [\mathbb{Z}^3 : \Lambda] = DE, \qquad E\mathbb{Z}^3 \subset \Lambda. \tag{B5}$$

The orbit $\mathcal{O} = \{GC : G \in \mathrm{SL}_3(\mathbb{Z}/h\mathbb{Z})\}$ satisfies

$$|\mathcal{O}| \ge h^8/(4D^3E^2). \tag{B6}$$

Indeed, the GL stabilizer of the diagonal matrix has its first column fixed, second column differing from e_2 by three elements annihilated by D, and third differing from e_3 by three elements annihilated by E. Its order is at most D^3E^3. Its determinant image contains all units $u \equiv 1 \pmod{h/E}$, whose proportion in the unit group is at least E/h, also for $E = h$. Moreover,

$$|\mathrm{GL}_3(\mathbb{Z}/h\mathbb{Z})| = h^9(1 - B^{-1})(1 - B^{-2})(1 - B^{-3}) \ge 21h^9/64.$$

Orbit-stabilizer and the determinant homomorphism give

$$\frac{|\mathrm{SL}_3(\mathbb{Z}/h\mathbb{Z})C|}{|\mathrm{GL}_3(\mathbb{Z}/h\mathbb{Z})C|} = \frac{|\det(\mathrm{Stab}_{\mathrm{GL}}(C))|}{|(\mathbb{Z}/h\mathbb{Z})^*|} \ge E/h,$$

proving (B6). Uniform G gives a uniform orbit element. All elements have the same row span and determinant as C.

3.4. **The auxiliary cap and its weights.** Choose a nonsquare $\nu \in \mathbb{F}_q$ and put

$$w = \left\lfloor \frac{q}{1000H^2} \right\rfloor, \qquad S = \{(x, y, x^2 - \nu y^2) : 0 \le x < w,\ y \in \mathbb{F}_q\}, \qquad s = wq.$$

The form $x^2 - \nu y^2$ vanishes only at $(0, 0)$. A line with nonvertical direction intersects its graph in at most two points, because the resulting quadratic has nonzero leading coefficient; a vertical line has one intersection. Thus S has no three distinct collinear points. Since $q \ge 2000H^2$,

$$s \ge q^2/(2000H^2), \qquad q^3/s \le 8000qh^2 < 2^{13}qh^2. \tag{B7}$$

Write $\|a\|_q$ for the absolute value of the least absolute integer representative, and define

$$A_{\text{allowed}} = \{a \in \mathbb{F}_q^3 : \|ma_1\|_q > 3Hw \text{ for } 1 \le m \le 3H\}.$$

Each multiplication by m is invertible modulo q. A uniform a violates its condition with probability $(6Hw+1)/q$. The union bound gives total at most $18/1000 + 3/(2000H) < 1/2$. Choose a uniformly from this allowed set and G_q uniformly from $\mathrm{GL}_3(\mathbb{F}_q)$, independently, and set $V = G_q(a + S)$. It has s elements, retains the cap property, and does not contain zero: otherwise $\|a_1\|_q \le w - 1$, contrary to the condition for $m = 1$.

No nonzero $x \in \mathbb{Z}^3$, $|x| \le H$, can satisfy $\sum_i x_i u_i = 0 \pmod q$ for three distinct $u_i \in V$. If $m = \sum_i x_i \ne 0$, then $1 \le |m| \le 3H$. Applying G_q^{-1}, the first coordinate gives $ma_1 = -\sum_i x_i v_{i,1}$ with $v_i \in S$. The right side has an integer representative of absolute value at most $3H(w-1)$, contradicting the allowed shift. If $m = 0$, the relation contradicts affine independence of distinct cap points; its coefficients are not all zero modulo q because $|x_i| \le H < q$.

For an integer matrix A define

$$W(A) = (q^3/s)^3 \mathbb{P}(\text{all columns of } A \pmod q \text{ belong to } V).$$

Then

$$\begin{aligned} &W(A) \le 4 && \text{for affinely independent columns modulo } q, \\ &W(A) \le 4(q^3/s) && \text{if the rank modulo } q \text{ is at least two}, \\ &W(A) \le 4(q^3/s)^2 && \text{always.} \end{aligned} \tag{B8}$$

For the first bound, before restricting a the affine map is uniform. There are $q^3(q^3-1)(q^3-q)$ ordered affinely independent triples, and $s(s-1)(s-2)$ in S; restricting a costs at most two. For the second, a specified independent pair has at most s^2 inverse pairs in $a + S$, among $(q^3-1)(q^3-q)$ pairs. For the third, a nonzero specified column has at most s inverse images among $q^3 - 1$ vectors; a zero column has weight zero. After multiplication by q^9/s^3, the stated constants follow since the denominators are at least half their leading powers for $q \ge 3$. Positive weight implies that no column is zero and that either the columns are affinely independent or an equal pair occurs. In the first case all nonzero integral relations of length at most H are excluded.

3.5. **Weighted singular and nonsingular counts.** Let the columns of A belong to

$$\mathcal{B} = ([0, N)^2 \times [N, 2N)) \cap \mathbb{Z}^3, \qquad \pi(u) = (u_1/u_3, u_2/u_3).$$

Fix C, Λ, I, D, E as above and an integer determinant value $|v| \le \tau$. For matrices with pairwise distinct projections and $A \pmod h \in \mathcal{O}$, we prove

$$\sum_A W(A) \le 2^{120} (\log_2(2N))^2 N^6 / I^2. \tag{B9}$$

If $v \ne 0$, then $|v| < q$ gives rank three modulo q, so $W \le 4$. Every row has length less than $4N$ and belongs to Λ, whose third successive length is at most $E \le h \le 4N$. Apply (B4) with $X = 4N$ and use $\log_2(8N) \le 3\log_2(2N)$. The resulting constant is below 2^{98}.

Suppose $v = 0$. Distinct projections imply rational rank exactly two, and the primitive integral null vector x has every coordinate nonzero. It is a cross product of independent rows divided by its coordinate gcd, so $|x| \le 16N^2$. Summing over both signs is an upper bound. Write

$$\Lambda_x = \Lambda \cap x^\perp, \quad x \cdot \Lambda = g(x)\mathbb{Z}, \quad J_x = \det \Lambda_x = I|x|/g(x).$$

By (B5), $g(x)$ divides E, and

$$g(x)^3/I \le E^2/D \le h^2. \tag{B10}$$

The lattice Λ_x reduces onto the whole plane $\Pi_x \subset \mathbb{F}_q^3$ perpendicular to $x \pmod q$. To see this, choose $z \in \mathbb{Z}^3$ with $x \cdot z = 1$. An integer lift u of a point of Π_x has $x \cdot u = qa$; replace it by $u - qaz$. This proves surjectivity for $\mathbb{Z}^3 \cap x^\perp$. Its multiple by E belongs to Λ_x and remains surjective since $q \nmid E$.

For an index-m sublattice $\Gamma \subset \Lambda_x$, the number of ordered row triples of length at most $4N$ spanning a rational plane is at most

$$2^{27}N^6/(mJ_x)^3. \tag{B11}$$

If there are any, two independent rows show that the second successive length is at most $4N$. The plane count gives at most $32(4N)^2/(mJ_x) = 2^9N^2/(mJ_x)$ choices per row.

For a dyadic shell $Y \le |x| < 2Y$ with $Y \ge h$,

$$\sum_{\substack{x \text{ primitive} \\ Y \le |x| < 2Y}} g(x)^3 \le 2^8 Y^3 I. \tag{B12}$$

For $j \le e$, $B^j \mid g(x)$ is equivalent to $Cx = 0 \pmod{B^j}$. The Smith form makes the fraction of such residue vectors $B^{-j-\max(j-b,0)}$. Since $Y \ge h \ge B^j$, each residue class has at most $125(Y/B^j)^3$ vectors in the ball of radius $2Y$. Using $g^3 \le \sum_{j=0}^{e} B^{3j}\mathbf{1}_{\{B^j \mid g\}}$, the sum is at most

$$125Y^3 \sum_{j=0}^{e} B^{2j-\max(j-b,0)} \le 250Y^3 I < 2^8 Y^3 I.$$

The last inequality follows because the exponents increase by at least one to $b + e$.

Partition the null vectors into shells $Y = 1, 2, 4, \ldots$. There are at most $5\log_2(2N)$ shells. For a fixed x, (B11) is bounded by

$$2^{27}N^6 g(x)^3/(m^3 I^3 Y^3). \tag{B13}$$

The support cases of (B8) are exhaustive:

(i) Affinely independent residues. Positive weight excludes $|x| \le H$, so $Y > H/2 = h$. Use (B12), (B13) with $m = 1$, and $W \le 4$. The shell contribution is at most $2^{37}N^6/I^2$.

(ii) An equal pair i, j, with $x \pmod q$ not proportional to $\delta = e_i - e_j$. The functional $v \mapsto v_i - v_j$ is nonzero on Π_x, so rows belong to an index-q sublattice of Λ_x. There are at most $125Y^3$ integer normals in the shell. Use (B10), (B13) with $m = q$, and the general weight bound. By (B7), one pair contributes at most

$$2^{62}(N^6/I^2)h^6/q \le 2^{62}N^6/I^2.$$

The three pairs contribute at most $2^{64}N^6/I^2$.

(iii) An equal pair with $x \pmod q$ proportional to δ. The coordinate outside the pair is a nonzero multiple of q, so $|x| \ge q$ and $Y > q/2$. There are $q - 1$ relevant residue classes, each with at most $(4Y/q + 1)^3 \le (6Y/q)^3$ vectors of length less than $2Y$. Thus there are at most $2^8Y^3/q^2$ normals. For rank at least two modulo q, (B8), (B10), (B13) with $m = 1$, and (B7) give at most $2^{50}(N^6/I^2)h^4/q \le 2^{50}N^6/I^2$ per pair. For rank at most one, the rows reduce into one of the $q + 1$ lines of Π_x, with zero row space covered by this union. Each inverse image has index q in Λ_x. Summing (B11) gives at most $2^{28}N^6/(q^2J_x^3)$ row triples. Together with the exceptional normal

count, (B10), and the general weight bound, this contributes at most $2^{64}(N^6/I^2)h^6/q^2 \le 2^{64}N^6/I^2$ per pair.

The sum of these cases, including the three equal pairs, is less than $2^{68}N^6/I^2$ per shell. Hence the zero-determinant count is at most $2^{71}\log_2(2N)N^6/I^2$, smaller than (B9). We counted all matrices with rows in Λ; imposing the orbit condition can only decrease this count. This proves (B9) also for singular C and every auxiliary rank pattern.

4. Sampling, deletion, and every cardinality

4.1. **The exact lifting distribution.** Independently choose G_h uniformly from $\mathrm{SL}_3(\mathbb{Z}/h\mathbb{Z})$ and the auxiliary random set V above. Conditional on these shared choices, each sample draws an independent uniform label and the independent digits of Section 2, giving a column $c \pmod h$; then it draws an independent uniform residue from V and a uniform lift $u \in \mathcal{B}$ with those residues modulo q and with $u = G_h c \pmod h$. Since $hq \mid N$, each prescribed residue pair has exactly $(N/(hq))^3$ lifts. All draws are fresh; only G_h, V are shared.

For a triple, condition on its labels and full digit matrix C. Its main residue is uniform on $\mathcal{O}$. The auxiliary probability of a prescribed triple is $W(A)/q^9$, including coincident columns. For any event $\mathcal{E}$ of lifts, the exact identity is

$$\mathbb{P}(A \in \mathcal{E} \mid C, \text{labels}) = \frac{h^9}{N^9|\mathcal{O}|} \sum_{\substack{A \in \mathcal{E} \\ A \bmod h \in \mathcal{O}}} W(A). \tag{B14}$$

No unconditional independence of samples is assumed.

For pairwise distinct projections with $|\det A| \le \tau$, the fixed determinant residue admits at most one integer determinant in this interval, because $2\tau < h$. It admits none unless $\mathcal{J}$ occurs. Combine (B6), (B9), and (B14) to bound the conditional probability by

$$2^{122}\big(\log_2(2N)\big)^2 \frac{hD}{N^3}\mathbf{1}_{\mathcal{J}}.$$

By (2), $\mathbb{E}[D\mathbf{1}_{\mathcal{J}}] \le 6/(Mr^5) + 2/r^{10} \le 8/r^8$. Thus the unconditional bad-triple probability is at most

$$2^{125}\big(\log_2(2N)\big)^2 h/(N^3 r^8). \tag{B15}$$

For two samples the equal-projection probability is at most

$$2^{20}(hq)^6/N^3. \tag{B16}$$

Equality of projections means proportional integer columns, since their third coordinates are positive. Every column is a positive integer multiple of a primitive z with $z_3 > 0$ and $|z| < 3N$. There are at most $3N/|z|$ multiples in $\mathcal{B}$. A dyadic shell $Y \le |z| < 2Y$ contains at most $125Y^3$ integer vectors. Summing the radii gives $\sum_{0<|z|<3N} |z|^{-2} \le 750N$, and hence at most $6750N^3$ ordered proportional pairs. Conditional on residues and shared choices, each lift is uniform on $(N/(hq))^3$ possibilities, and the two lifts are independent. Divide the absolute pair count by $(N/(hq))^6$ to obtain (B16).

4.2. **Deletion with an explicit threshold.** Set $a_r = r^\gamma\sqrt{N^3/\tau}$, $n_r = \lfloor a_r \rfloor$, and draw $2n_r$ samples. Let Z count equal-projection index pairs and bad index triples with pairwise distinct projections and determinant of absolute value at most τ. Linearity of expectation and (B15)–(B16) give

$$\frac{\mathbb{E}Z}{n_r} \le 2^{21}\frac{n_r(hq)^6}{N^3} + 2^{126}\frac{n_r^2\big(\log_2(2N)\big)^2 h}{N^3 r^8}. \tag{B17}$$

Counting the events requires no independence between them. Using $\tau \ge B^{k-1}/3$ and $B > r^6$, the first term is at most

$$2^{22}r^\gamma B^{(3s_0-k+1)/2} = 2^{22}r^\gamma B^{-102195/2} < 2^{22}r^{\gamma-306585} < 1/8.$$

This estimate retains the slack needed for the smaller lift box.

For the second term write $x = \log_2 r$. Since $k < 2^{19}$ and $80000k^4 < 2^{93}$, $\log_2 B < 93 + 6x$. Moreover,

$$N \le 16B^{28k-s_0}, \qquad \log_2(2N) \le 5 + 28k(93+6x) < 701kx < 2^{29}x$$

for $r \ge 37$. As $h/\tau \le 3B$, the second term of (B17) is at most

$$2^{279}x^2 r^{2\gamma-2} \le 2^{300}x^2 2^{-\delta x}, \qquad \delta = 2/(\rho+1) > 2^{-26}.$$

For $r \ge R$, $x \ge 2^{50}$. The function $x^2 2^{-\delta x}$ is decreasing there since $2/x < \delta \log 2$. At $x = 2^{50}$ the last bound is less than $2^{400-2^{24}} < 1/8$. Consequently $\mathbb{E}Z < n_r/4 < n_r$, so an outcome with $Z < n_r$ exists.

For every original violating pair or triple mark one of its indices. Removing all marked indices deletes at most Z samples and meets every violation. At least n_r samples remain; retain exactly n_r. Their projections are distinct points in $[0,1)^2$ and every triple has $|\det A| > \tau$, including exclusion of determinant zero. For retained columns the exact projective identity is

$$\mathrm{Area}(\pi(u),\pi(v),\pi(w)) = \frac{|\det(u,v,w)|}{2u_3v_3w_3} \ge \frac{\tau}{16N^3}.$$

The ordinary area factor is included and all denominators belong to $[N, 2N)$. Since $a_r \ge 2$ and $n_r \ge a_r/2$, the retained set P_r satisfies

$$n_r^2 \Delta(P_r) \ge r^{2\gamma}/64. \tag{B18}$$

4.3. **Effective transfer to every cardinality.** Put $\beta = (83k - 3s_0 + 1)/2 = \rho/6$. The bounds on q and τ give

$$\sqrt{2}\, r^\gamma B^\beta < a_r \le 64\sqrt{3}\, r^\gamma B^\beta.$$

Combining the interval for B and $n_r \ge a_r/2$ yields

$$r^\alpha \le n_r \le D_0 r^\alpha, \qquad D_0 = 2^{20\rho}. \tag{B19}$$

For the lower bound, $40000k^4 \ge 4$ and $\beta \ge 1$ give $a_r \ge 2r^\alpha$. For the upper bound, $64\sqrt{3}(80000k^4)^\beta < 2^{7+93\rho/6} < 2^{20\rho}$. These bounds hold for every allowed $r \ge R$, without monotonicity of n_r or of the choices of B, q.

Now fix any integer $n \ge N_0$ and let $y = n^{1/\alpha}$. Since $\alpha < \rho + 1$, $y \ge R$. Section 3.1 supplies a prime $r \equiv 1 \pmod 3$ in $(y, 2y]$. Hence

$$n \le n_r \le C_0 n, \qquad C_0 = 2^{21\rho+1},$$

where C_0 bounds $D_0 2^\alpha$. Delete excess points of P_r; minimum triangle area cannot decrease on taking a subset. Using (B18), (B19), and $2\gamma/\alpha = 2/(\rho+2) = \eta$, the resulting n-point set satisfies

$$\Delta(P) \ge \frac{n^{-2+\eta}}{64 D_0^\eta C_0^{2-\eta}} \ge 2^{-100\rho} n^{-2+\eta}.$$

Indeed, $0 < \eta < 1$ makes the denominator at most $64 D_0 C_0^2 = 2^{62\rho+8} < 2^{100\rho}$.

All choices can be made by finite enumeration: order field presentations and choose the first irreducible polynomial of degree five, the first allowed roots and primes, and ordered digit outcomes and matrices. Exhaustively enumerate the finite sampling space and take the first outcome with fewer than n_r violations, whose existence follows from (B17). Mark the least index in each violation, remove the union of marked indices, and retain the first n points. No computational practicality is asserted. Finally, $100866719 < 2 \cdot 50893237$, proving the asserted strict comparison and completing Theorem 1. □

5. Exact finite ingredients

The mathematical supplement checks the eighteen-form polynomial identity, reserved-word packing and exponent arithmetic. The geometric and all-cardinality arguments are proved in the preceding sections. The construction specifies c, N_0 and a finite search for every $n \geq N_0$.

References

1. Michael A. Bennett, Greg Martin, Kevin O'Bryant, and Andrew Rechnitzer, *Explicit bounds for primes in arithmetic progressions*, Illinois Journal of Mathematics **62** (2018), no. 1-4, 427–532, The proof uses arXiv:1802.00085v3, Theorem 1.2 and equations (1.10)–(1.11).
2. Ivan Blinov, *Heilbronn triangle bounds: finite packings and improved scales*, GitHub research draft, 2026, Pinned checkpoint e29a2912; retrieved October 17, 2026.
3. Austin Conner, Fulvio Gesmundo, Joseph M. Landsberg, and Emanuele Ventura, *Rank and border rank of Kronecker powers of tensors and Strassen's laser method*, arXiv:1909.04785v2, 2021, Final preprint version, Theorem 4.1.
4. OpenAI, *A power improvement in the Heilbronn triangle lower bound*, OpenAI Math Release preprint, September 25, 2026, Pinned source commit 1581d4cb.

Mathematical supplement

The exact arithmetic programs are included in the attached mathematical supplement (ZIP). Open the attachment in a PDF reader with file-attachment support.